# Neerav Poluri

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## Education

**Purdue University,** West Lafayette, IN — Aug. 2025 – May 2028
*Bachelor of Science in Computer Science* — GPA: 3.82

## Technical Skills

- *Languages:* C++, Python, TypeScript, JavaScript, SQL, C, Java, Rust, C#
- *Software:* Linux, Git, CMake, React, Bun, SQLite, WebAssembly, Three.js, NumPy, OpenCV
- *Robotics:* ROS 2, Isaac Sim, SLAM, Path Planning, Reinforcement Learning, PID Control
- *CAD/Manufacturing:* SolidWorks, Fusion 360, Onshape, CFD, DFM, FDM 3D Printing, CNC Machining

## Projects

**FDM Print-Time Estimator** — Sept. 2026 – Present

- Developed a C++17 print-time estimation library with TypeScript bindings for STL/STEP models and OrcaSlicer printer profiles
- Reused object toolpaths across packing rotations, achieving a 5.53× speedup over OrcaSlicer in cached six-part plate benchmarks
- Built a reusable WebAssembly library for model orientation, support visualization, and approximate print-time estimates
- Validated cache reuse, profile changes, and worker error recovery through C++ unit and native integration tests

**High-Speed 3D Printer Development (Cuprite)** — Mar. 2022 – Present

- Designed and built high-performance 3D printers, including Cuprite, a compact CoreXY printer designed in Fusion 360
- Experimentally characterized printer behavior and used test data to guide system modifications and optimization
- Applied CAD, DFM, CFD, and rapid prototyping while working with sponsors to source components

**Robotic 3-Axis Gantry** — Apr. 2026

- Designed and built a three-axis gantry integrating mechanical components and motion-control systems
- Engineered an underactuated gripper to accommodate objects with varied geometry using fewer actuators
- Developed OpenCV image-processing pipelines and integrated YOLO for real-time object detection and localization

## Experience

**Hub for Innovation, Ventures, and Entrepreneurship (HIVE)** – *Community Engagement Specialist* — Oct. 2026 – Present

- Designed makerspace layouts and reorganized equipment for a joint HIVE and Purdue Hackers facility
- Secured company sponsorships and sourced fabrication equipment for the makerspace
- Established procedures for member access and equipment use

**Mechanisms and Robotics Systems Lab** – *Undergraduate Researcher* — Sept. 2026 – Present

- Collecting manipulation-task data using a Franka Panda arm, roller gripper, GelSight Mini, and custom Python scripts
- Researching pan-tilt-zoom (PTZ) cameras for visual perception and data collection in world-action models

**Purdue Hackers** – *Engineering Organizer* | West Lafayette, IN — Aug. 2026 – Present

- Implemented browser-based model preparation in React and TypeScript for PrintHive, a 3D-print farm management system
- Added email-confirmed print cancellation and desktop/mobile push notifications backed by SQLite transactions
- Organizing and running Hack Nights with the Purdue Hackers team

**Bechtel Innovation Design Center** – *Boilermakery TA* — Oct. 2025 – Oct. 2026

- Improved and maintained Linux-based slicing, print-data logging, and backup systems supporting 10,000+ student print hours per semester
- Repaired and upgraded FDM printers to improve reliability and support reservation-free student printing
- Led workshops on design for manufacturing and FDM print optimization

**Setlurs.com & Quasar Design** – *Edge Computing & AI Intern* — Sept. 2024 – May 2025

- Built computer-vision and machine-learning systems for automated 3D-print monitoring and process control
- Designed an automated print-removal system integrating computer vision and mechanical systems